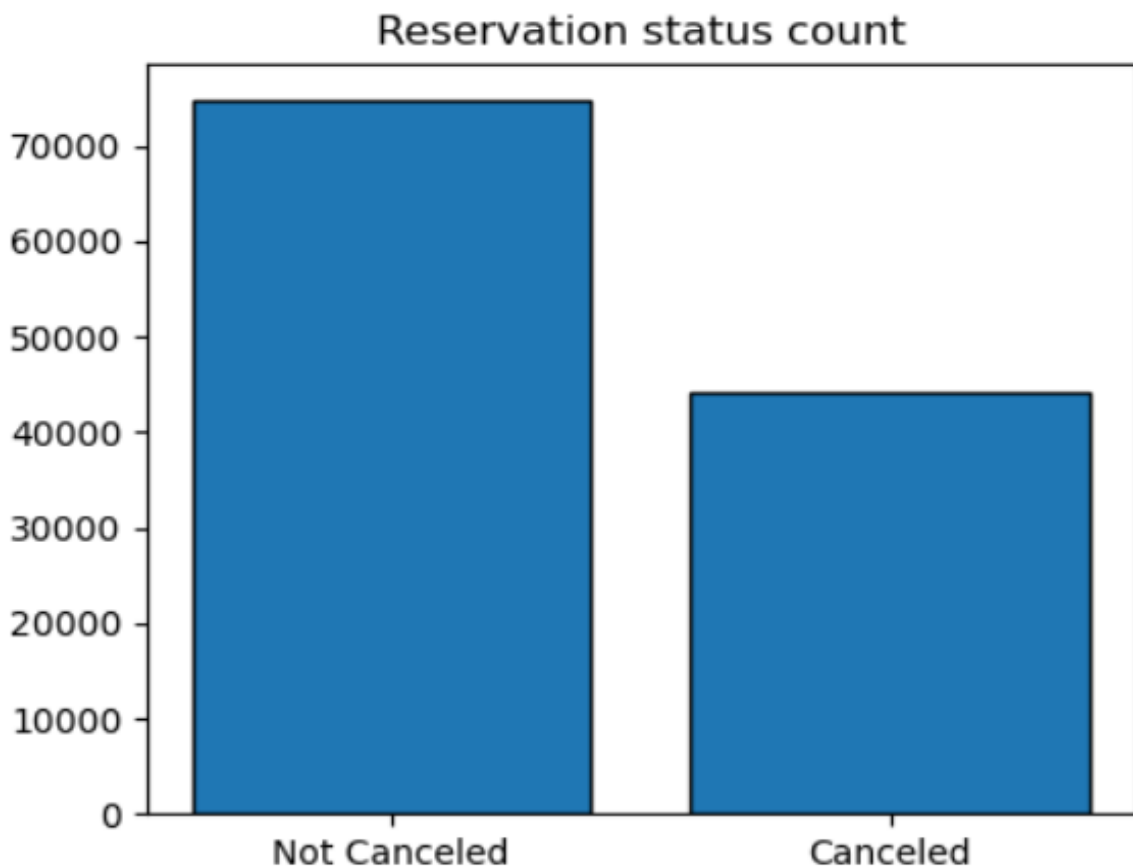


Results & Recommendations

Report

```
is_canceled  
False    0.628653  
True     0.371347  
Name: proportion, dtype: float64
```



Upon reviewing the data and examining the figures shown in the chart above, we found that the booking cancellation rate reached 37%, representing more than one-third of all bookings—during the recent period. This necessitates a deeper analysis to explore the potential causes and underlying issues.

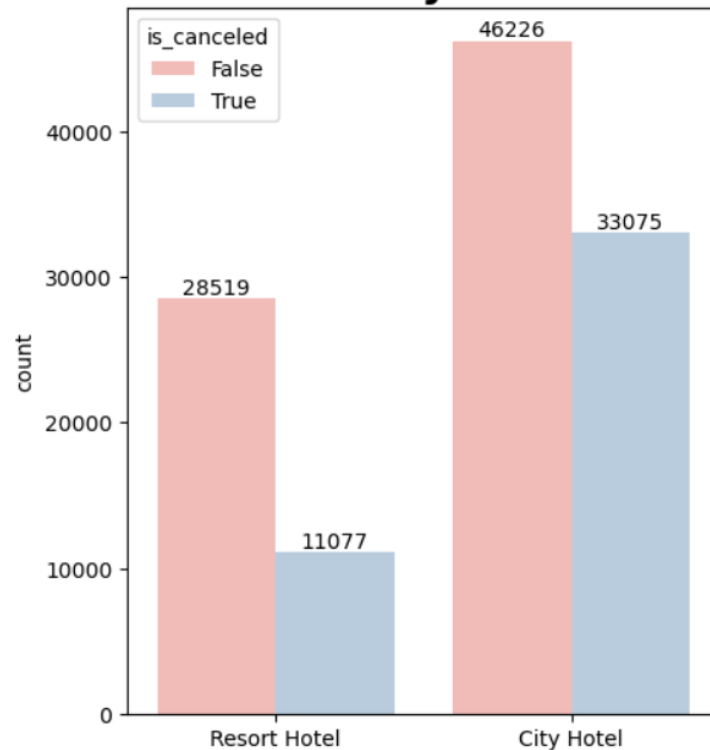
City Hotel Cancellation distribution:

is_canceled	
False	0.582918
True	0.417082

Resort Hotel Cancellation distribution:

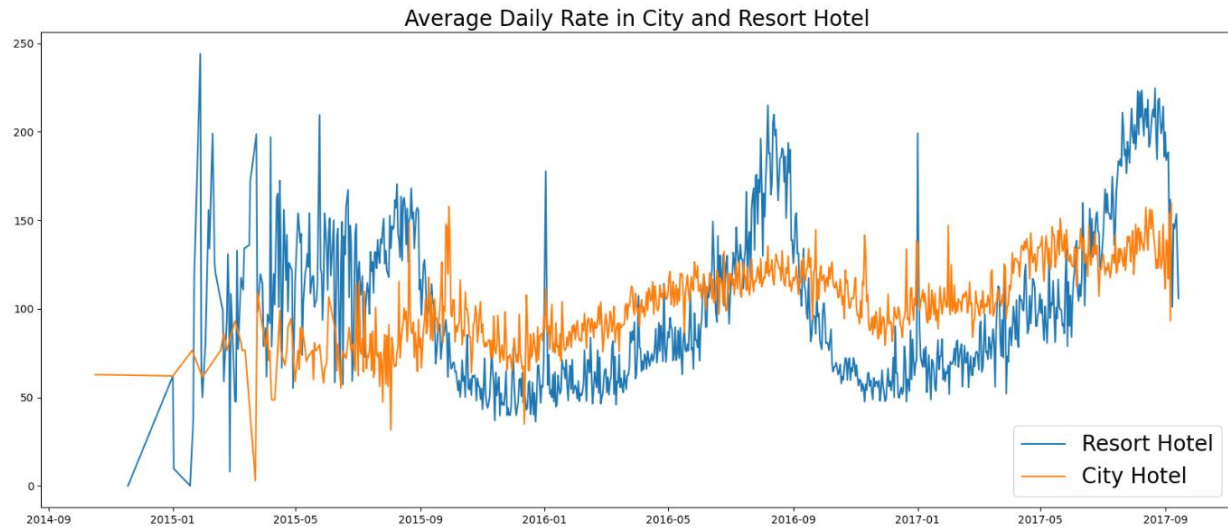
is_canceled	
False	0.72025
True	0.27975

Cancellation rates in City hotel and Resort hotel



Upon reviewing the booking data, we found that the City Hotel had the highest number of completed bookings; however, the Resort Hotel boasted a lower cancellation rate approximately 28% compared to roughly 42% for the City Hotel.

In comparison to resort hotels, city hotels have more bookings. Its possible that resort hotels are more expensive that those in cities.

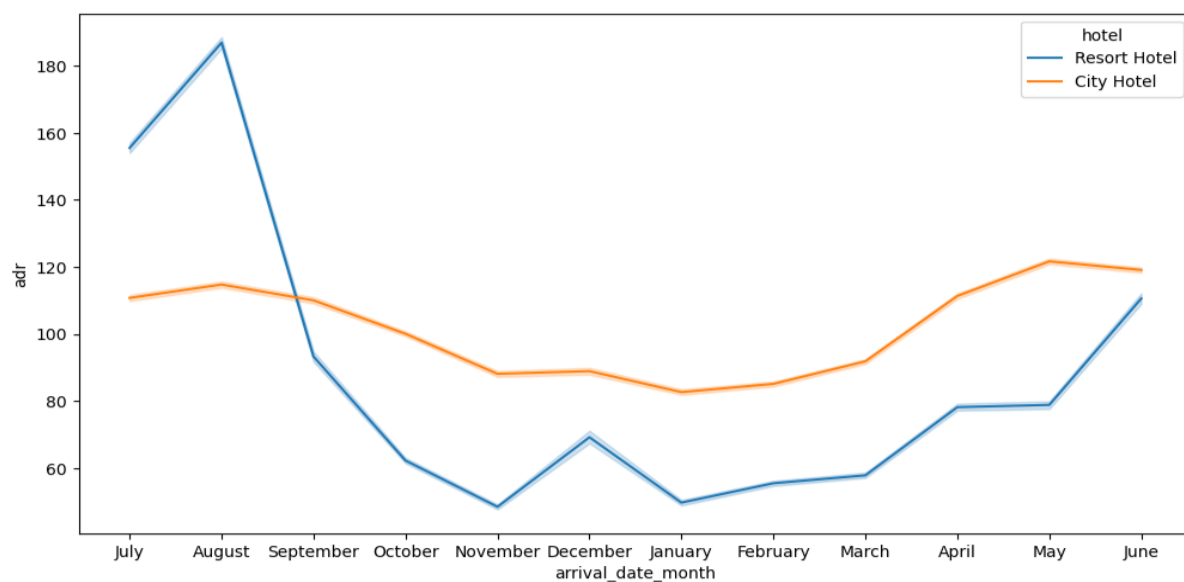


Resort hotel rates are significantly higher.

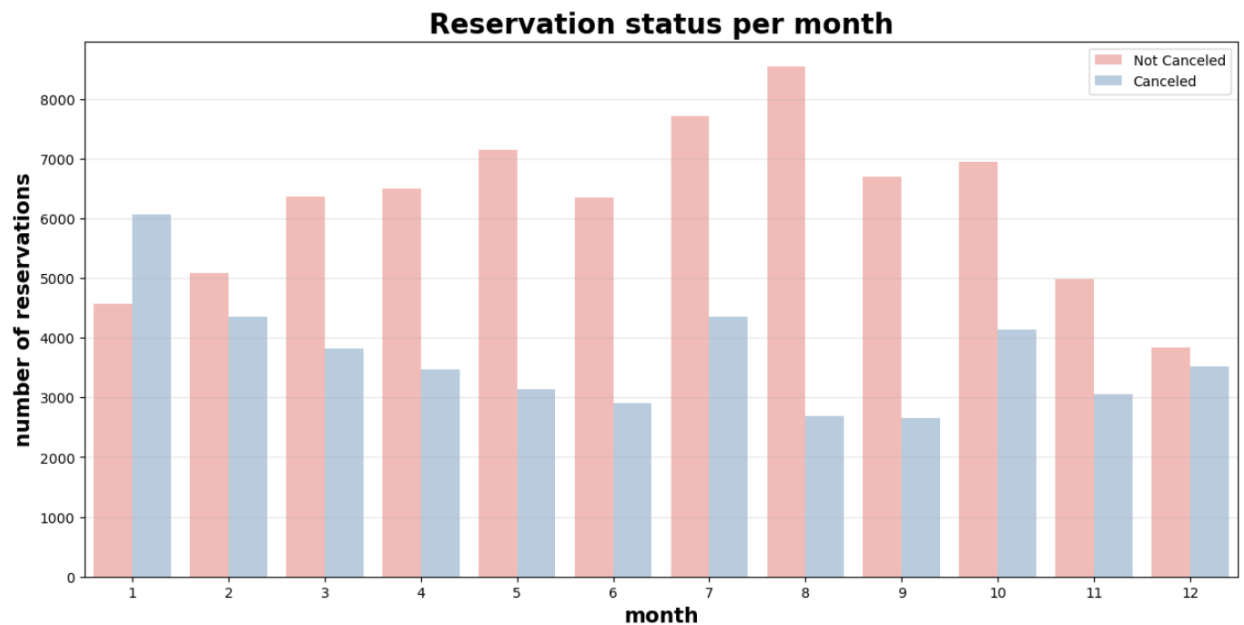
In contrast, city hotel rates do not experience major fluctuations.

The chart above shows that the average daily rate for a city hotel is—on certain days lower than that of a resort hotel, and drops even further on other days.

Needless to say , Resort hotel rates may rise during weekends and public holidays.



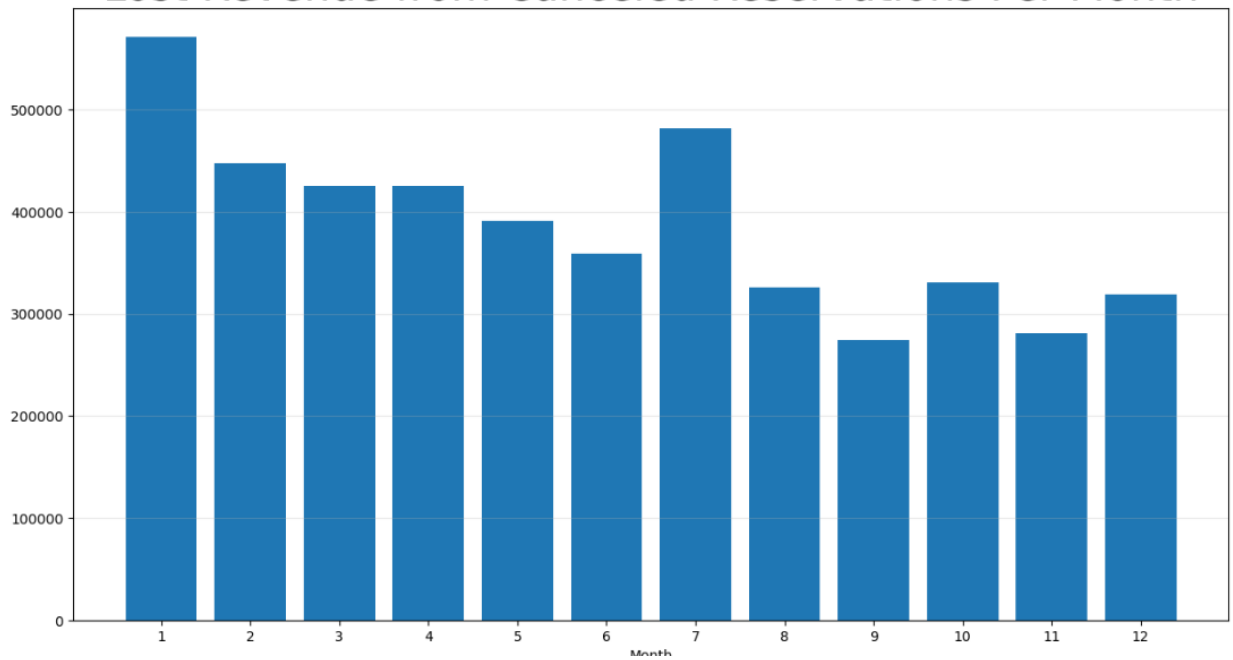
For Resort Hotel, ADR is more expensive during July, August & September and for City Hotel, ADR is slightly more during March, April & May.



Most bookings were done in the month of May, June, July, August.

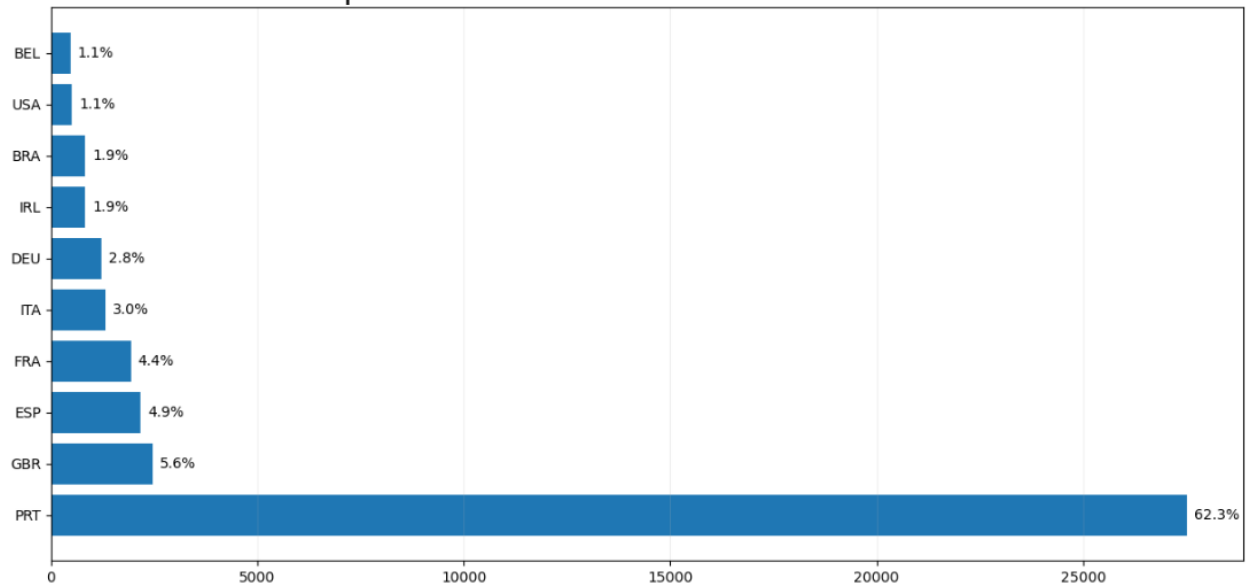
We've created a grouped bar graph to examine the months with the highest and lowest reservation levels based on their status. It's evident that the month of August stands out, having the highest numbers of both confirmed and canceled reservations. In contrast, January has the fewest confirmed reservations but the highest number of canceled reservations.

Lost Revenue from Canceled Reservations Per Month



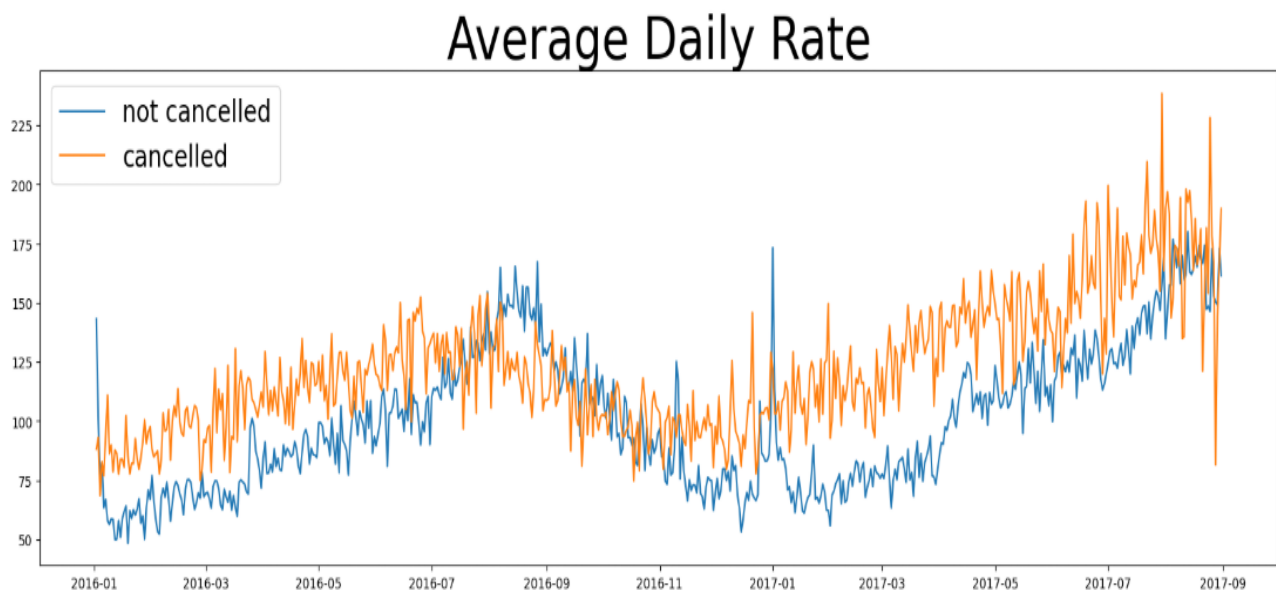
This bar chart illustrates that January and July are primarily responsible for the highest percentage of lost profits due to booking cancellations.

Top 10 countries with reservation canceled



```
country
PRT    0.566306
GBR    0.202243
FRA    0.185694
ESP    0.254085
DEU    0.167147
ITA    0.353956
IRL    0.246519
BEL    0.202391
BRA    0.373201
NLD    0.183935
Name: is_canceled,
```

While Portugal accounts for 62.3% of all cancellations, this is partly explained by Portugal also being the largest source market. More importantly, guests from Portugal show a markedly higher **cancellation rate** (56.6%) compared to other top source countries (16–37%) indicating a genuine behavioral pattern, not just a volume effect. This is consistent with domestic guests having more flexible, lower-commitment booking habits (e.g., no upfront payment, easier to cancel without financial loss) compared to international travelers who've already committed to flights and travel plans.



The graph shows cancellations are more common during periods of higher average daily rates, suggesting price could be a contributing factor to cancellation behavior though seasonality likely plays a role as well.

Suggestions:

- Increasing prices are associated with a higher rate of cancellations. To mitigate reservation cancellations, hotels could refine their pricing strategies by offering reduced rates for specific locations and providing discounts to customers.
- We also realise that the high rate of cancellations can be due to high no deposit policies.
- Introduce a partial non-refundable deposit or stricter cancellation window specifically for domestic (Portuguese) bookings, especially during high-demand months.
- Monitor cancellation rates by country over time to confirm the pattern holds and isn't seasonal.
- Consider offering early-bird incentives that reward guests for confirming/locking in bookings rather than general advertising campaigns, since the issue is post-booking commitment, not booking volume.
- During the month of January, hotels can launch marketing campaigns with attractive offers to boost their revenue, especially since cancellations tend to peak during this period.
- The resort hotel experiences a lower ratio of cancellations compared to the city hotels.